Operator dependence and the rank of Γ - AI-Generated note

Introduction

The quantity your algorithm drives down on a cut is rank(Γ), with Γ the matrix of your Eq. (4). The sets U and V behind its rows and columns are formed by removing duplicated operator chains, so their elements are distinct *as symbols*. The identity rank(Γ) = minimal bond needs more than that. It needs the chains to be linearly independent *as operators*, and that is a property of the matrices the symbols name, which the elimination does not hold. Where it fails, the bond is above the operator-Schmidt rank of the operator and no elimination on Γ reaches it. A Hamiltonian meets this whenever its local vocabulary is dependent. An XY chain whose hopping is written with ladder operators and whose anisotropy is written with Cartesian components comes out at bond 6 on every cut where the operator-Schmidt rank is 4. The smallest instance of the same failure is a controlled gate whose target carries a multiple of the identity.

This note states the condition exactly, gives the counterexample, and adds a pre-processing step that establishes the condition over the rationals before Γ is built. The step is exact and leaves the operator unchanged. It returns a Hamiltonian whose operators are already independent untouched, which includes every sum of Pauli strings and so every Hamiltonian of the experiments in your Sec. IV A.

1 When rank(Γ) is the minimal bond

Fix one cut. With the notation of your Sec. III A, writing $n = |U|$ and $m = |V|$,

$$\hat{O} \;=\; \sum_{i=1}^{n}\sum_{j=1}^{m} \Gamma_{ij}\, \hat{U}_i \otimes \hat{V}_j. \tag{1}$$

Let D_U and D_V be the total physical dimensions of the two sides of the cut, let vec stack the entries of an operator in a fixed order, and collect the chains in

$$L = \Big[\operatorname{vec}\hat{U}_1 \quad \cdots \quad \operatorname{vec}\hat{U}_n\Big] \in \mathbb{C}^{D_U^2 \times n}, \qquad R = \Big[\operatorname{vec}\hat{V}_1 \quad \cdots \quad \operatorname{vec}\hat{V}_m\Big] \in \mathbb{C}^{D_V^2 \times m}. \tag{2}$$

Reshaping $\hat{O}$ across the cut into the matrix M whose row index is the pair of physical indices on the left and whose column index is the pair on the right, eq. (1) reads

$$M \;=\; L\,\Gamma\,R^{\mathsf{T}}. \tag{3}$$

The smallest bond this cut admits, in any MPO or TTNO that represents $\hat{O}$, is rank(M). Together with eq. (3) this gives

$$\operatorname{rank}(M) \;\le\; \operatorname{rank}(\Gamma), \tag{4}$$

with equality when L and R have full column rank, because multiplication by a matrix with independent columns on either side leaves the rank unchanged. Full column rank of L is exactly the statement that $\hat{U}_1, \ldots, \hat{U}_n$ are linearly independent operators. The Frobenius rank inequality bounds the gap by the two dependence defects,

$$\operatorname{rank}(M) \;\ge\; \operatorname{rank}(\Gamma) - (n - \operatorname{rank} L) - (m - \operatorname{rank} R). \tag{5}$$

The condition is decided one node at a time. Suppose every node carries a linearly independent set of operators. The chains of U are then distinct tensor products of elements of those sets, hence

distinct elements of a basis of the tensor product of their spans, hence linearly independent, so L has full column rank. The same holds for R. Conversely, if the chains of U are dependent, then some node in the left part carries a dependent set, since otherwise the previous sentence applies. So

$$\text{operators independent on every node} \iff \operatorname{rank}(\Gamma) = \operatorname{rank}(M) \text{ on every cut}, \tag{6}$$

and the right side holds for every cut at once or fails only at cuts separated by a node that carries a dependence. The argument uses only that a cut splits the nodes into two sets, so it covers a tree exactly as it covers a chain. One consequence is a test that costs nothing. A node whose p distinct symbols are matrices of dimension d carries a dependence as soon as $p > d^2$. The converse does not hold, since three operators of dimension two can already be dependent, so $p \le d^2$ screens nothing and a Hamiltonian is immune only when the operators it draws on are themselves independent.

Your paper states a condition of this shape for the bipartite method of Ren et al. [2], on the prefactors rather than on the operators, *"This algorithm can only find the optimal bond dimension if all prefactors γ_i of Eq. (3) are different and independent"* (p. 4), and for SGE it reports *"We found that our SGE-based algorithm achieves optimal bond dimensions and outperforms the original purely bipartite-graph-based algorithm of [11] for a diverse set of Hamiltonians. A mathematical proof of optimality is still outstanding"* (p. 15). The condition in eq. (6) is on the operators and is independent of the prefactors. It is not established by the elimination, which stores the entries of Γ as pairs (c, s_i) and prohibits the sum of two different symbols by your Eq. (8), so combining two dependent operators into one is outside the moves available to it.

2 A spin chain over a dependent vocabulary

Take the chain

$$\hat{H} \;=\; \sum_i J(\hat{S}_i^{+}\hat{S}_{i+1}^{-} + \hat{S}_i^{-}\hat{S}_{i+1}^{+}) \;+\; \sum_i g(\hat{X}_i\hat{X}_{i+1} + \hat{Y}_i\hat{Y}_{i+1}) \;+\; \sum_i h\,\hat{Z}_i, \tag{7}$$

with $J = 1$, $g = 0.4$ and $h = 0.3$. The hopping is written with ladder operators and the anisotropy with Cartesian components, which is what happens when the two usual forms of the same chain are both present in one Hamiltonian. An interior node then carries the six symbols $\hat{S}^{+}$, $\hat{S}^{-}$, $\hat{X}$, $\hat{Y}$, $\hat{Z}$, $\hat{I}$, and $\hat{S}^{\pm} = (\hat{X} \pm i\hat{Y})/2$, so four of them span two dimensions. The counting test of section 1 also flags the node, $p = 6 > d^2 = 4$.

On six sites the construction gives bond 6 on every cut, where the operator-Schmidt rank of $\hat{H}$ is 4. So eq. (4) is strict, by the dependence defect of the node vocabulary. Nothing in the input marks the case. Every string is distinct, the prefactors are distinct, and the chain sets U and V are non-redundant as symbols. Rewriting the four dependent symbols over two of them, as in section 4, brings every cut to 4 and takes the number of strings from 26 to 16.

3 A controlled gate, the minimal instance

The same failure at the smallest size it occurs. A controlled gate enters as the three strings of $\hat{C}_b\hat{U} = \hat{I} + \hat{\Pi}_b \otimes (\hat{U} - \hat{I})$, namely $\hat{\Pi}_b \otimes \hat{U}$ with prefactor $+1$, $\hat{\Pi}_b \otimes \hat{I}$ with -1, and $\hat{I} \otimes \hat{I}$ with $+1$, where $\hat{\Pi}_b$ is the projector onto the control pattern. Take a target that carries a phase on both of its levels, $\hat{U} = \lambda\hat{I}$. A controlled phase gate in the usual sense, $\hat{U} = \operatorname{diag}(1, e^{i\theta})$, is not this case, since that matrix is independent of $\hat{I}$ and the gate already comes out at its rank.

Take the gate on the left of fig. 1, four qubits with controls on q_1 and q_2 and the target on q_4, and the cut between q_2 and q_3. The three strings give $U = \{\hat{\Pi}_b,\ \hat{I} \otimes \hat{I}\}$ and $V = \{\hat{I} \otimes \hat{U},\ \hat{I} \otimes \hat{I}\}$, both



Figure 1: Two controlled phases. On the left two controls next to the target, on the right a gap between the controls. The target carries $\lambda\hat{I}$, which is dependent on the $\hat{I}$ the other two strings carry there.

non-redundant as symbols, and

$$\Gamma = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \qquad \operatorname{rank}(\Gamma) = 2. \tag{8}$$

No row or column operation on eq. (8) lowers that. The two elements of V are $\hat{I} \otimes \hat{U} = \lambda\, \hat{I} \otimes \hat{I}$, so $\operatorname{rank} R = 1$, and eq. (3) gives $\operatorname{rank}(M) = 1$. The gate is supported on its controls alone, $\hat{C}_b\hat{U} = \hat{I} + (\lambda - 1)\, \hat{\Pi}_b \otimes \hat{I}$, and one bond state carries it. The built bonds are $[2, 2, 2]$ where the operator-Schmidt rank is $[2, 1, 1]$.

4 Rewriting over an independent set

On node k let the distinct symbols be $s_1, \ldots, s_p$ with matrices $A_1, \ldots, A_p$. Choose $i_1 < \cdots < i_q$ so that $A_{i_1}, \ldots, A_{i_q}$ is a basis of the span of all of them, and write each operator in that basis,

$$A_t \;=\; \sum_{b=1}^{q} c_{tb}^{(k)}\, A_{i_b}, \qquad t = 1, \ldots, p. \tag{9}$$

Substituting eq. (9) into a term of your Eq. (3) and expanding the product over the nodes,

$$\gamma \bigotimes_k A_{t_k} \;=\; \sum_{b_1, \ldots, b_N} \gamma \Big(\prod_k c_{t_k b_k}^{(k)} \Big) \bigotimes_k A_{i_{b_k}}, \tag{10}$$

and strings that have become equal are collected into one with the sum of their prefactors. The operator is unchanged, every node now carries only the operators of its own basis, and these are independent by construction, so eq. (6) holds and $\operatorname{rank}(\Gamma)$ is the minimal bond on every cut. Nodes whose operators are already independent have $q = p$ and $c_{tb}^{(k)} = \delta_{tb}$ in eq. (9), so they contribute one summand to eq. (10) and nothing moves. A Hamiltonian independent on every node is returned as the object it came in as.

Exactness. A prefactor is carried as $a + ib$ with a and b rational. A float converts to a rational without rounding, being a dyadic rational already, so no conversion loses a digit. The elimination that produces eq. (9) takes the first nonzero entry of a column as its pivot, with no comparison of magnitudes, which is sound because the arithmetic is exact. No rounding error can therefore report a dependence that is not there or miss one that is, and the bond is a property of the operator rather than of the floating point format.

The symbol set. A prefactor that stays rational after eq. (10) is returned on the symbol it came in with. Only one that turns complex is given a symbol of its own, and equal values share it. The set S therefore grows, before the elimination runs, by at most the number of distinct complex prefactors the rewrite creates, and the restriction of your Eq. (8) during the elimination is untouched.

Term count. Equation (10) is a product over the nodes, so the rewrite can produce more strings than it consumed, and collection can leave fewer. Both directions occur, and both are measured below.

5 When an exact dependence must be refused

Over the rationals a set of operators is dependent or it is not, with no tolerance in between. That is the property that makes the bond exact, and it is also a trap. Operators that are dependent only to within rounding are exactly independent, and operators that are exactly dependent on a basis that is nearly rank deficient have coordinates in eq. (9) of size $\|A_t\|/\sigma_q$ with σ_q the smallest singular value of the basis. Evaluating $\sum_b c_{tb} A_{i_b}$ in double precision then loses the operator to cancellation.

Five diagonal operators of dimension four, four of which agree to 10^{-9}, have exact rank four and numerical rank three. The singular values of the first four are 4.15, $5.3 \cdot 10^{-9}$, $3.2 \cdot 10^{-16}$ and $5.3 \cdot 10^{-17}$, the exact coordinates of the fifth on them reach $4.8 \cdot 10^{16}$, and evaluating eq. (9) in double precision misses the operator by 11.4 times its largest entry.

A dependence found exactly is therefore used only when it also holds in double precision. It must rebuild the operator to a relative 10^{-10}, and the kept operators must have numerical rank q at the same tolerance. A node that fails either test is left as it is, which costs a bond of the TTNO and never the operator it represents. This is what keeps a generator whose entries are floats, such as a Lindbladian, correct under the pass.

6 What changes

Every number below is produced by `experiments/independent_operators/independent_operators_probe.py` and saved in `probe_output.txt` beside it. Each case is built twice, once with the rewrite and once without, the two dense operators are compared, and the rank column is the operator-Schmidt rank from a singular value decomposition of that operator. Bonds are listed left to right along the chain.

Table 1: Cases in which the bond falls. The controlled phases carry $\lambda = e^{0.7i}$ and the sweep behind the first three rows runs one to three controls and two to eight sites, nineteen cases, every one of which lands on the rank.

case	without the rewrite	with it	rank
one control, 8 sites	$2,2,2,2,2,2,2$	$1,1,1,1,1,1,1$	$1,1,1,1,1,1,1$
two controls, 8 sites	$2,2,2,2,2,2,2$	$2,1,1,1,1,1,1$	$2,1,1,1,1,1,1$
three controls, 8 sites	$2,2,2,2,2,2,2$	$2,2,1,1,1,1,1$	$2,2,1,1,1,1,1$
controls on 1 and 4, 8 sites	$2,2,2,2,2,2,2$	$2,2,2,1,1,1,1$	$2,2,2,1,1,1,1$
controlled identity, 6 sites	$2,2,2,2,2$	$1,1,1,1,1$	$1,1,1,1,1$
$\hat{I}\hat{X} - \hat{P}_0\hat{X} - \hat{P}_1\hat{X} + \hat{P}_0\hat{Z}$	2	1	1
$\hat{n}\hat{n}$ coupling with a $\hat{Z}$ field, 6 sites	$3,3,3,3,3$	$2,3,3,3,2$	$2,3,3,3,2$
XY chain, ladder and Cartesian, 6 sites	$6,6,6,6,6$	$4,4,4,4,4$	$4,4,4,4,4$
Lindbladian, two jumps, 2 qubits	$5,4,5$	$4,4,4$	$4,4,4$
Lindbladian, two jumps, 3 qubits	$5,4,6,4,5$	$4,4,6,4,4$	$4,4,5,4,4$

The first four rows are the gate of section 3, and the bond is 1 on every cut outside the span of the controls, which is where the gate acts.

The rows that matter in practice are the next three, where the dependence comes from the vocabulary a Hamiltonian is written in rather than from a gate. The seventh row is a chain with coupling $\hat{n} \otimes \hat{n}$ and field $\hat{Z}$, where $\hat{n} = (\hat{I} - \hat{Z})/2$, so the three symbols $\hat{n}$, $\hat{Z}$ and $\hat{I}$ on a node span two dimensions. The eighth is an XY chain whose hopping is written with $\hat{S}^+$ and $\hat{S}^-$ and whose anisotropy is written with $\hat{X}$ and $\hat{Y}$, four symbols on a node spanning two dimensions, and its string count falls from 26 to 16 as well. Writing one chain in two vocabularies is ordinary,

and nothing warns the author that it costs half again on every bond. The last two rows are the symbolic Lindbladian of `generate_lindbladian` with two jump operators per qubit, on a chain whose nodes are the ket and the bra copy of each qubit.

Table 2: Cases that do not move, which is the argument that the rewrite is safe to run by default.

case	result
randomised Hamiltonians over $\{\hat{X}, \hat{Y}, \hat{Z}, \hat{I}\}$ on the eight-site tree of your Sec. IV A, 30 terms, 20 seeds	the same object is returned 20 times out of 20, so the TTNO is identical by construction
`MultiControlledGate` with $\hat{X}$, $\hat{Y}$, $\hat{Z}$ and $\hat{H}$ on four qubits	bonds $2, 2, 2$ either way, the same object is returned
a Lindbladian in the Pauli basis with float entries, three and four qubits	the TTNO matches the dense generator to $2.2 \cdot 10^{-15}$ and $3.6 \cdot 10^{-15}$, the node being refused by the test of section 5
controlled gates on four qubits whose target is $\mathrm{diag}(1, e^{i\theta})$, $\hat{X}$ or $\hat{H}$	bonds $2, 2, 2$ either way, equal to the rank, and the terms are returned unchanged, those targets being independent of $\hat{I}$
one term, and a sum in which every prefactor cancels	the pass is the identity and does not return an empty Hamiltonian

The first row is general rather than fortunate. Those Hamiltonians draw their operators from four matrices of dimension two that form a basis, so every node carries an independent set and eq. (6) already holds. No Hamiltonian built from a pool of independent operators, and in particular no sum of Pauli strings, can be affected by the rewrite in either direction, and no second diagram is built for one.

What the rewrite does not guarantee. Equation (6) makes $\mathrm{rank}(\Gamma)$ equal to the minimal bond. It does not make the elimination attain $\mathrm{rank}(\Gamma)$, and the rewrite changes the presentation the elimination runs on, so the rewritten terms can come out worse. On the three-qubit chain of the last row with one jump operator per qubit instead of two, the terms as given reach 5 on the middle cut, which is the rank, while the terms rewritten reach 6. Lines 20 and 21 of your Algorithm 1 keep the better of two results in the same situation, and the pass does the same. Both diagrams are built whenever the rewrite changes the terms, and the one with the smaller total bond dimension is kept, so no case measured here comes out above the bond the terms as given would have produced. What remains is that neither presentation need attain $\mathrm{rank}(\Gamma)$. The middle cut of the last row of table 1 stands at 6 against a rank of 5.

Cost. The rewrite is exact rational arithmetic on p vectors of length d^2 per node, and it ran in under 0.04 s on every case above. A Hamiltonian it leaves alone costs one diagram, as before, and one it changes costs two, which came to between 0.005 s and 0.11 s in total for every case in table 1. The string count can fall or rise. A controlled gate goes from three strings to two, the chain of the seventh row of table 1 from 17 to 12, and the Lindbladian of the last row from 34 to 38, since eq. (10) is a product over the nodes. The pass is reached through `combine_dependent_operators`, which defaults to `True`, and passing `False` restores the previous path exactly.

References

[1] H. Çakır, R. M. Milbradt and C. B. Mendl, *Optimal Symbolic Construction of Matrix Product Operators and Tree Tensor Network Operators*, arXiv:2502.18630.

[2] J. Ren, W. Li, T. Jiang and Z. Shuai, *A general automatic method for optimal construction of matrix product operators using bipartite graph theory*, J. Chem. Phys. **153**, 084118 (2020).